

Application-aware quantum resource compiler (AQRC) for fault-tolerant molecular energy calculations

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Abstract

Fault-tolerant quantum chemistry pays for error correction per logical qubit and per non-Clifford rotation. Current resource estimates assign one code distance to every logical qubit and one synthesis precision to every rotation, both set from a single global error target. This note places the accuracy target on the observable instead. Each error location in the circuit is assigned the sensitivity of the observable to an error there, and the two quantities a surface-code compiler can vary, the distance of each patch and the synthesis precision of each rotation, are allocated from that sensitivity. We derive the structure of the sensitivity tensor for molecular Hamiltonians. The occupation-flipping (X type) sensitivity of a spin-orbital equals its orbital energy at the mean-field level. The phase-flip (Z type) sensitivity vanishes on a single determinant and is bounded by the occupation fluctuation of the spin-orbital. The first-order response of the energy to any coherent error is bounded by twice the energy standard deviation of the prepared state and vanishes on an eigenstate. The physical layer is generated and validated with Stim and PyMatching rather than taken from the literature. On LiH and on a stretched H_4 chain, at chemical accuracy and a physical error rate of 10^{-3} , the sensitivity spread is below the suppression factor of one distance step and the allocation returns uniform distances. Two effects of the Jordan–Wigner encoding are identified. Errors inside a flattened excitation generator are amplified in proportion to $\sin^2$ of the string angle, and the X sensitivity of a qubit depends on its position in the Jordan–Wigner ordering through the parity string.

1 Question and scope

A fault-tolerant calculation of a molecular energy is judged by one number. The energy must lie within chemical accuracy, 1 kcal/mol or 1.6 mHa, of the exact value of the chosen model. Resource estimates for such calculations set the code distance from a logical error target that is uniform over the circuit [1, 2, 3]. The target is a proxy. What is required is that the final observable be accurate.

Not every logical error moves the energy by the same amount. We call the amount the sensitivity of the observable to the error. If sensitivities differ across logical qubits or across error types, protection can be redistributed. The same reasoning was used earlier to distribute measurement shots over the terms of a Hamiltonian [4].

Three questions have to be answered. Which quantities can a fault-tolerant compiler actually change. How large are the sensitivities and where do they concentrate for molecular Hamiltonians. How much does the redistribution save once idle time and the coupling between patches are included. Sections 2 to 7 answer the first and the third question with a model that is explicit at every step. Section 8 answers the second question for two small molecules with exact simulation.

AQRC, application-aware quantum resource compiler, is a compilation pass. Its input is a program in the Pauli-rotation instruction set of Section 2.1 and an observable. Its output is a protection specification, the distance of every patch and the synthesis precision of every rotation together with the error budget they realise. It does not perform layout, scheduling or gate synthesis, which belong to the compilers and estimators that consume the specification. The word compiler is used in the sense in which register allocation is part of a classical compiler. The code is built on Stim and PyMatching [5, 6]. Every quantity used below is defined once in the definitions document of the repository, with the function that implements it.

Resource estimators such as Qualtran [23], the Azure Quantum Resource Estimator [3], pyLIQTR [24] and BenchQ [25] map logical counts to physical resources under one global error budget, and a recent study optimises how that global budget is split between logical errors, magic states and synthesis [26]. Partial error correction protects some logical qubits and leaves others unprotected [28]. AQRC produces a budget resolved per patch and per rotation, upstream of the estimators. Hardware-side sensitivity analysis of the kind embedded in IonQ’s Walking Cat architecture [29] asks which physical channel dominates the logical error; the question here is which logical error dominates the observable. The time and cost models of Sections 5 and 6 are specific to lattice surgery on a planar surface code and are kept in their own module, so that a machine with transversal gates and qubit transport can be described by replacing them while the sensitivity tensor stays the same.

2 Setting

2.1 Instruction set

The circuit is a sequence of Pauli product rotations

$$U = U_K \cdots U_1, \quad U_k = \exp(-i\theta_k P_k), \quad (1)$$

acting on n logical qubits prepared in a computational basis state $|\Phi\rangle$. Here P_k is a Pauli string, θ_k a real angle and K the number of rotations. Under the Jordan–Wigner mapping [7] $|\Phi\rangle$ is the Hartree–Fock determinant and the rotations are the exponentiated excitation generators of an adaptive derivative-assembled pseudo-Trotter variational quantum eigensolver (ADAPT-VQE) ansatz [8]. Each fermionic generator maps to a set of mutually commuting Pauli strings, so its exponential is a product of rotations of the form (1) without Trotter error. The angles are optimised once and then fixed. Everything below is a property of the fixed circuit.

The form (1) is the instruction set of fault-tolerant surface-code computation. In Pauli-based computation with lattice surgery [9, 10, 11] a program is compiled into a sequence of Pauli product rotations. Clifford rotations are commuted to the end and tracked in software. Each remaining rotation by an arbitrary angle is synthesised into a sequence of $\pi/8$ Pauli product rotations. Each $\pi/8$ rotation is executed as one multi-patch Pauli product measurement between the data patches in the support of P_k and a magic-state patch, followed by a classically controlled Clifford correction. A gate of the original circuit does not correspond to a physical operation. The natural unit of logical time is one $\pi/8$ rotation. The set of patches involved in rotation k is the support of P_k , denoted S_k .

2.2 What the architecture can vary

A surface-code compiler exposes a small number of parameters. The dimensions of each patch, which may differ between patches. The dimensions of a patch as a function of time, changed by code deformation at a cost of about one patch distance in rounds. The number of syndrome rounds per Pauli product measurement. The precision of each synthesised rotation and the quality of the magic

states it consumes. The code distance of a gate is not on this list. During a multi-patch measurement the merged region contains a logical operator that passes through the smallest participating patch, so the protection of every participant is limited by the smallest distance among them.

Heterogeneous distances are already standard inside magic state factories. The 15-to-1 protocols of Litinski use different distances for different qubits and stages of the same protocol, chosen by how errors at each stage propagate to the output [12]. AQRC applies the same reasoning at the level of the application. Two parameters carry the allocation, the distance d_i of the patch holding logical qubit i and the synthesis precision ϵ_k of rotation k .

2.3 Error budget

Let δ_O be the accuracy target on the observable O . For the energy $\delta_O = 1.6$ mHa. It is split as

$$\delta_O = \delta_{\text{ans}} + \delta_{\text{mem}} + \delta_{\text{syn}} + \delta_T. \quad (2)$$

δ_{ans} is the ansatz error, the difference between the ideal circuit energy and the exact energy of the model. δ_{mem} bounds the logical errors during the computation. δ_{syn} bounds the synthesis error. δ_T bounds the error of the magic states, and is computed from the factory output error and the sensitivities on the support of each rotation. The split between δ_{mem} and δ_{syn} is optimised in Section 7.4.

3 Physical layer

3.1 Simulation

Logical error rates are generated for the code, the decoder and the noise structure in use. We simulate rotated surface-code memory experiments with Stim and decode with the minimum-weight perfect matching decoder of PyMatching [5, 6]. Circuit-level noise places a depolarising channel after every Clifford gate, data depolarisation before every round, and flip errors before measurement and after reset. The uniform structure sets all four channels to the same rate p . The memory experiment that protects the logical Z observable counts logical X errors, and the experiment that protects logical X counts logical Z errors. Both are run.

A memory experiment with r rounds returns a per-shot failure probability $P(r)$. The failures of successive rounds cancel in pairs, and initialisation and readout add a boundary term, so

$$P(r) = \frac{1}{2} [1 - (1 - 2\epsilon_b)(1 - 2\epsilon)^r], \quad (3)$$

with ϵ the per-round logical error rate and ϵ_b the boundary term. Each experiment is run at $r = d, 2d$ and $3d$, and ϵ is the slope of $\ln(1 - 2P)$ against r . The inset of Figure 1a shows the three points for $d = 5$, $p = 0.2\%$. The conversion from a single $r = d$ experiment, $1 - (1 - P)^{1/d}$, underestimates ϵ at $d = 7$ by 10 to 40%, because the time boundaries of a short experiment are better protected than its bulk.

3.2 Fit and validation

Below threshold the per-round logical error rate follows

$$p_L(d, p) = A (p/p_{\text{th}})^{(d+1)/2}. \quad (4)$$

Here d is the patch distance, p the physical error rate, p_{th} the threshold and A a prefactor. The form goes back to the empirical scaling of Fowler et al. [13]. The exponent is the minimum number

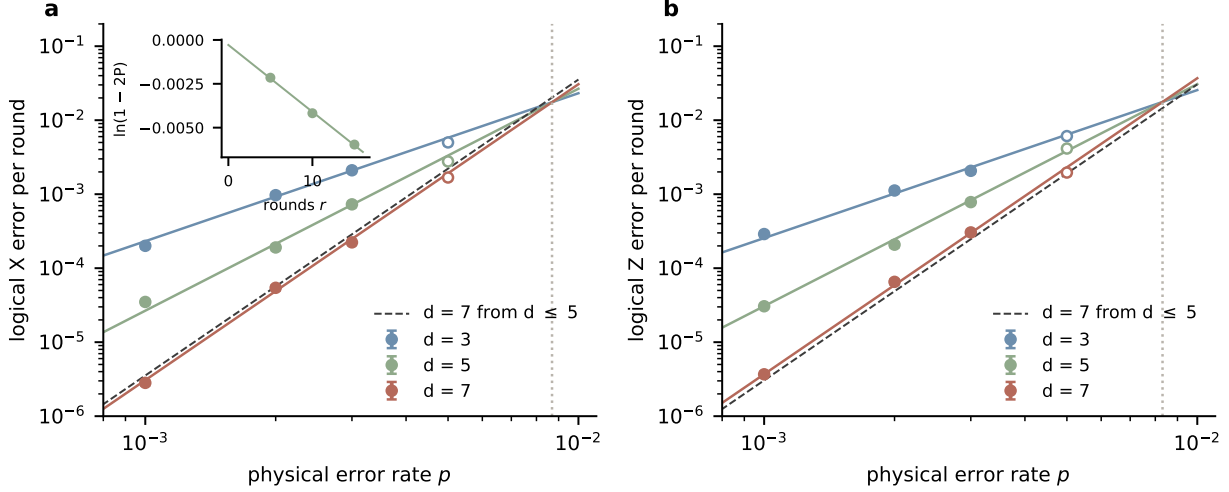


Figure 1: Physical layer. Per-round logical error rate of the rotated surface code under uniform circuit-level noise, decoded with PyMatching, for logical X (a) and logical Z (b). Points are the slope of $\ln(1 - 2P)$ over $r = d, 2d, 3d$; error bars are one standard deviation from the binomial counts. Solid lines are the fit (4) to the filled points, $p \leq 0.3\%$. Open points at $p = 0.5\%$ are not in the fit. The dashed line is $d = 7$ predicted from a fit to $d \leq 5$. The dotted vertical line is the fitted p_{th} . Inset in (a), the three round counts for $d = 5$, $p = 0.2\%$ and the fitted slope.

of physical faults that a minimum-weight decoder can misread as a logical operator of length d . The same form is the default logical error model of the Azure Quantum Resource Estimator [3], and it is equivalent to the suppression factor $\Lambda = p_L(d)/p_L(d+2) = p_{\text{th}}/p$ used in the scaling experiments of Google Quantum AI [14, 15].

Taking logarithms of (4) gives a relation linear in the two unknowns $\ln A$ and $\ln p_{\text{th}}$, fitted by least squares separately for X and Z logical errors on the points with $p \leq 0.3\%$, where p/p_{th} is below 0.4 and the power law holds. Only points with at least ten observed errors in every round count are used. Figure 1 shows the data and the fit. For logical X the fit gives $A = 0.0175$ and $p_{\text{th}} = 0.87\%$, for logical Z it gives $A = 0.0176$ and $p_{\text{th}} = 0.83\%$, with residuals of 0.054 and 0.040 in $\log_{10}$. Both thresholds lie in the range known for this code and decoder under circuit-level noise. The fit is validated before it is used. Fitting $d \leq 5$ alone and predicting $d = 7$ reproduces the held-out points within 0.11 (X) and 0.13 (Z) in $\log_{10}$, the dashed lines in the figure. The points at $p = 0.5\%$, open symbols, fall below the power law and bunch together as the three distances approach threshold. Molecular workloads need d between 15 and 25 [1, 2], so the model is extrapolated in d ; at $p = 10^{-3}$ and $d = 13$ it gives 4.7×10^{-9} (X) and 6.4×10^{-9} (Z) per round, and $\Lambda = 8.7$ and 8.3.

4 Application layer

4.1 Definition of the sensitivity

Let $|\psi\rangle = U|\Phi\rangle$ be the ideal output and O the observable. For an error Q , a single-qubit Pauli on qubit i , inserted after t rotations, the corrupted output is

$$|\psi'_{t,i,Q}\rangle = U_K \cdots U_{t+1} Q U_t \cdots U_1 |\Phi\rangle. \quad (5)$$

If the error occurs with probability p the output is the mixture $(1-p)|\psi\rangle\langle\psi| + p|\psi'\rangle\langle\psi'|$ and the observable becomes

$$\langle O \rangle_p = \langle \psi | O | \psi \rangle + p (\langle \psi' | O | \psi' \rangle - \langle \psi | O | \psi \rangle), \quad (6)$$

exactly, for one error location. We define the sensitivity as

$$s_{t,i,Q}(O) = |\langle \psi'_{t,i,Q} | O | \psi'_{t,i,Q} \rangle - \langle \psi | O | \psi \rangle|, \quad (7)$$

the coefficient of the error probability in the observable, with Q ranging over X , Y and Z . When several locations ℓ fail independently with probabilities p_ℓ , the observable shifts by $\sum_\ell p_\ell \Delta_\ell$ plus terms of second order in the probabilities, where Δ_ℓ is the signed version of (7). The first-order bound

$$|\langle O \rangle - \langle \psi | O | \psi \rangle| \leq \sum_\ell p_\ell s_\ell \quad (8)$$

is the error functional used for allocation. Computing the tensor costs $3n(K+1)$ partial circuit simulations, a few seconds for $n \leq 8$ by state-vector simulation. The tensor is kept with all three indices.

4.2 Structure of the sensitivity for molecular Hamiltonians

Write the Hamiltonian as a Pauli sum $H = \sum_\alpha c_\alpha P_\alpha$. Fix a Pauli string Q and split $H = H_c + H_a$ into the terms that commute with Q and the terms that anticommute with Q .

Proposition 1. *For any state $|\psi\rangle$, $\langle \psi | Q H Q | \psi \rangle - \langle \psi | H | \psi \rangle = -2\langle \psi | H_a | \psi \rangle$.*

Proof. $Q H_c Q = H_c$ and $Q H_a Q = -H_a$. □

An error at the end of the circuit, $t = K$, has $|\psi'\rangle = Q|\psi\rangle$, so Proposition 1 gives its sensitivity directly. For an earlier error the same statement holds with H replaced by the Heisenberg-evolved operator $U_{>t}^\dagger H U_{>t}$, which is why the tensor is computed numerically. In the code the identity at $t = K$ agrees with the numerical tensor to 6×10^{-15} Ha.

Under the Jordan–Wigner mapping qubit i carries spin-orbital i and the number operator is $n_i = (1 - Z_i)/2$. A Pauli term contains X_i or Y_i exactly when the corresponding fermionic term changes the occupation of spin-orbital i .

Proposition 2 (phase errors). *Let $|\Phi\rangle$ be a single Slater determinant in the orbital basis of the mapping. Then a Z_i error applied to $|\Phi\rangle$ does not change the energy, for every i . For a general state $|\psi\rangle$ the energy change caused by Z_i is $-2\langle \psi | H_a^{(i)} | \psi \rangle$, where $H_a^{(i)}$ collects the terms in which spin-orbital i appears exactly once.*

Proof. The terms anticommuting with Z_i are exactly the terms containing X_i or Y_i , that is the terms that change the occupation of i . Their expectation in a determinant vanishes because a determinant is an occupation eigenstate. Proposition 1 gives the general expression. □

Corollary 1 (occupation-fluctuation bound). *Let $n_i = \langle \psi | a_i^\dagger a_i | \psi \rangle$ and let W_i be the sum of the absolute coefficients of the fermionic terms of H in which spin-orbital i appears exactly once. Then*

$$s_{Z_i}(H) \leq 2 W_i \min(\sqrt{n_i}, \sqrt{1-n_i}) \leq 2 W_i \sqrt{2 n_i (1-n_i)}. \quad (9)$$

Proof. Each term of $H_a^{(i)}$ has the form $c a_i^\dagger A$ or $c A' a_i$, with A, A' products of ladder operators on other spin-orbitals and $\|A\| \leq 1$. By Cauchy–Schwarz, $|\langle \psi | a_i^\dagger A | \psi \rangle| \leq \|a_i \psi\| \|A \psi\| \leq \sqrt{n_i}$. Anticommuting A to the other side of $a_i^\dagger$ gives instead $\leq \|a_i^\dagger \psi\| = \sqrt{1 - n_i}$. The same holds for the terms with a_i . The triangle inequality over the terms gives the first inequality and $\min(\sqrt{n}, \sqrt{1 - n}) \leq \sqrt{2n(1 - n)}$ on $[0, 1]$ the second. $\square$

The bound vanishes on a determinant and grows with the occupation fluctuation of the spin-orbital. In the weak-correlation limit the off-diagonal elements of the one-body density matrix are first order in the excitation amplitudes while $n_i(1 - n_i)$ is second order, so s_{Z_i} is expected to scale as $\sqrt{n_i(1 - n_i)}$.

Proposition 3 (occupation errors). *Let $|\Phi\rangle$ be the Hartree–Fock determinant in canonical orbitals of the active-space Fock operator, and let ε_i be the orbital energy of spin-orbital i . In the mean-field limit, where the circuit is the identity and the error acts on $|\Phi\rangle$, the energy change caused by X_i or by Y_i has magnitude $|\varepsilon_i|$.*

Proof. X_i flips the occupation of spin-orbital i , and Y_i does the same up to a phase. The result is the determinant with one electron added to an empty orbital or removed from an occupied one, all other orbitals unchanged. With frozen orbitals the energy difference is $+\varepsilon_i$ for addition and $-\varepsilon_i$ for removal. This is Koopmans’ theorem in its determinant form [16]. It holds exactly for the active-space Hamiltonian because the core contribution is folded into the one-body integrals exactly. $\square$

For a correlated state the occupation flip also removes the correlation energy that spin-orbital i participated in, so $s_{X,i}$ exceeds $|\varepsilon_i|$ by an amount of the order of the correlation energy. Propositions 2 and 3 together predict the shape of the tensor. Across qubits the X sensitivity follows the orbital energies. Across types the ratio of X to Z sensitivity is large for weakly correlated systems and approaches one as correlation grows. Section 8 confirms both predictions and shows the second one failing for a stretched H_4 chain.

Remark 1 (position dependence). *X_i on qubit i is not the fermionic operator $a_i + a_i^\dagger$. It is that operator times the Jordan–Wigner parity string of the qubits below i . The string of the second qubit of a spin-orbital pair contains the Z of its partner. The X sensitivity of a qubit therefore depends on its position in the ordering whenever the partner occupation fluctuates, while the Z sensitivity, which carries no string, does not. Section 8.2 measures this.*

4.3 Two observables

The energy is the right observable when it is read out as an expectation value. If the prepared state is the input of phase estimation, the relevant loss is the reduction of the overlap $F = |\langle \psi_0 | \psi \rangle|^2$ with the target eigenstate $|\psi_0\rangle$, since F is the success probability of the estimation [17]. Relation (6) holds for F because F is the expectation value of a projector. Both tensors are computed. An error that changes the particle number leaves the N -electron sector and destroys the overlap regardless of the orbital, while its effect on the energy is bounded by an orbital energy.

4.4 Concentration diagnostic

For a non-negative vector v of length N , with $N_{\text{eff}} = (\sum_j v_j)^2 / \sum_j v_j^2$, the concentration is

$$\eta = 1 - N_{\text{eff}}/N. \quad (10)$$

$\eta = 0$ means all entries are equal. It is evaluated along each axis of the tensor separately, so that the axis carrying the structure is identified before any allocation is attempted.

5 Time model and idle errors

The memory experiments of Section 3 measure a patch that sits still. Their per-round rate is the error rate of an idle patch. Rotation k is synthesised into $n_T(\epsilon_k)$ rotations by $\pi/8$, each a Pauli product measurement on S_k repeated for as many rounds as the largest participating distance, so its duration in code cycles is

$$r_k = n_T(\epsilon_k) d_{\text{time}}(k), \quad d_{\text{time}}(k) = \max_{j \in S_k} d_j. \quad (11)$$

During these rounds a patch in S_k is part of the merged region and its protection is that of the smallest participant. A patch outside S_k is idle and keeps its own distance,

$$d_{\text{eff}}(k, i) = \min_{j \in S_k} d_j \quad \text{if } i \in S_k, \quad d_{\text{eff}}(k, i) = d_i \quad \text{otherwise.} \quad (12)$$

The state is not defined inside a synthesised rotation, so the sensitivity of an error that occurs while rotation k executes is taken as the larger of the values just before and just after it, $s_{k,i}^Q = \max(s_{k-1,i,Q}, s_{k,i,Q})$. The memory error functional is

$$\text{Err}(d, \epsilon) = \sum_{k=1}^K r_k \sum_{i=1}^n \left[s_{k,i}^X p_{LX}(d_{\text{eff}}(k, i)) + s_{k,i}^Z p_{LZ}(d_{\text{eff}}(k, i)) \right] \leq \delta_{\text{mem}}. \quad (13)$$

Logical Y errors are dropped because a simultaneous logical X and Z failure is of second order. The idle share is the part of (13) with $i \notin S_k$.

Three couplings follow. A patch of low sensitivity idles for the whole circuit, so its exposure is $\sum_k r_k$ times its own rate and its distance cannot be reduced without limit. The duration of every step is set by the largest participating patch. A higher synthesis precision increases n_T , lengthens the circuit and increases the idle error, so δ_{mem} and δ_{syn} are not independent.

6 Distance allocation

6.1 Problem

The cost is the space-time volume of the data patches and the total volume,

$$V_{\text{data}} = \sum_k r_k \sum_i d_i^2, \quad V_{\text{total}} = (1 + \alpha) V_{\text{data}} + c_T N_T, \quad N_T = \sum_k n_T(\epsilon_k), \quad (14)$$

in qubit-rounds, where α is the ancilla and routing overhead of the data block and c_T the volume of the factory per magic state. We use $\alpha = 0.5$, the compact block of Litinski [11], and $c_T \approx 1.9 \times 10^5$ qubit-rounds, a 15-to-1 factory at $p = 10^{-3}$ with output error $p_T = 4.5 \times 10^{-8}$ [12]. The allocation problem is

$$\min_{d \in D^n} V_{\text{total}}(d, \epsilon) \quad \text{subject to} \quad \text{Err}(d, \epsilon) \leq \delta_{\text{mem}}, \quad (15)$$

with D the set of odd distances considered.

6.2 The distance gap rule

A continuous relaxation shows what to expect. Suppose the patches were decoupled, each idle for the whole circuit with total exposure $R = \sum_k r_k$ rounds, so that the error of patch i is $s_i R A \Lambda^{-(d_i+1)/2}$ with s_i its total sensitivity. The Lagrangian of (15) with cost $\sum_i d_i^2$ is stationary when $d_i \Lambda^{(d_i+1)/2} = \lambda' s_i$ with λ' common to all patches. For two patches this gives

$$d_1 - d_2 \approx \frac{2 \ln(s_1/s_2)}{\ln \Lambda}, \quad (16)$$

where a term $2 \ln(d_1/d_2)/\ln \Lambda$ has been dropped as small for the distances in question. A sensitivity ratio of Λ is needed to move a distance by one step of 2. The saving of one step on a patch is $1 - (d-2)^2/d^2$, about 28% at $d = 13$. The total saving is bounded by the spread of the sensitivities measured in units of Λ . This spread is computed before the solver is run.

6.3 Baselines, controls and solver

The uniform baseline gives every patch the smallest common distance that satisfies (13). The type-agnostic baseline allocates per patch but replaces both sensitivities by $\max_Q s_{k,i,Q}$, which is what an allocation without type resolution has to do. Two controls test whether an allocation, when it differs from uniform, comes from the structure of the tensor. The shuffled control permutes the qubit labels of the tensor, allocates, and evaluates the result on the true tensor. The mean-field control allocates from $s_{X,i} = |\varepsilon_i|$ and $s_{Z,i} = 0$, the content of Propositions 2 and 3, and evaluates on the true tensor. The solver enumerates D^n when the space has fewer than 3×10^5 points and otherwise descends greedily from the uniform solution with single-patch and pair moves; on small instances the exhaustive optimum checks the greedy result.

7 Rotation synthesis allocation

7.1 T-count model

A rotation by an arbitrary angle is approximated to precision ϵ by a Clifford+ T sequence with

$$n_T(\epsilon) = \lceil a \log_2(1/\epsilon) + b \rceil, \quad (17)$$

$a = 3$ and $b = 0$ for the Ross-Selinger algorithm [18]. Fallback and probabilistic protocols lower the slope [19]. Every $\pi/8$ rotation consumes one magic state and one time step of (11).

7.2 Error model, stationarity and coherent errors

In the angle picture a synthesis error of size ϵ_k on rotation k shifts the energy by at most

$$\Delta E_k(\epsilon_k) = g_k \epsilon_k + \frac{1}{2} h_k \epsilon_k^2, \quad g_k = \left| \frac{\partial E}{\partial \theta_k} \right|, \quad h_k = \left| \frac{\partial^2 E}{\partial \theta_k^2} \right|. \quad (18)$$

Proposition 4 (stationarity). *If the angles θ minimise $E(\theta) = \langle \Phi | U^\dagger H U | \Phi \rangle$, then $g_k = 0$ for every k and the energy is quadratic in every on-axis synthesis error.*

Proof. A minimiser of a differentiable function is a stationary point. □

A synthesised rotation differs from its target by a unitary whose generator need not be P_k . The next statement covers every direction.

Proposition 5 (coherent errors). *Let a coherent error $\exp(-i\varepsilon G)$, with G Hermitian, be inserted after t rotations, and let $E(\varepsilon)$ be the energy of the output. Then*

$$\left. \frac{dE}{d\varepsilon} \right|_{\varepsilon=0} = i\langle\psi|[G_t, H]|\psi\rangle, \quad G_t = U_{>t} G U_{>t}^\dagger, \quad (19)$$

and $|dE/d\varepsilon| \leq 2\|G\| \sigma_H$ with $\sigma_H^2 = \langle\psi|H^2|\psi\rangle - \langle\psi|H|\psi\rangle^2$. If $|\psi\rangle$ is an eigenstate of H the derivative vanishes for every G and every t .

Proof. Differentiating $\langle\Phi|U_{\leq t}^\dagger e^{i\varepsilon G} U_{>t}^\dagger H U_{>t} e^{-i\varepsilon G} U_{\leq t}|\Phi\rangle$ at $\varepsilon = 0$ gives (19). The bound is the Robertson uncertainty relation $|\langle[A, B]\rangle| \leq 2\sigma_A\sigma_B$ with $\sigma_{G_t} \leq \|G\|$. For an eigenstate with eigenvalue E_0 , $\langle\psi|G_t H - H G_t|\psi\rangle = E_0\langle G_t\rangle - E_0\langle G_t\rangle = 0$. $\square$

Remark 2 (directions that matter). *For a number-conserving Hamiltonian and circuit, a single-qubit direction X_i or Y_i has zero first-order response, because $H_t|\psi_t\rangle$ stays in the N -electron sector while $G|\psi_t\rangle$ leaves it. If state and Hamiltonian are real, as here, Z_i has zero first-order response too, since the commutator of two real symmetric operators has zero expectation in a real state. The directions with a first-order response are the number-conserving generators, the space of excitation operators, and the check of Section 8.1 is performed in that space.*

Writing a synthesised rotation as $V_k = U_k \exp(-i\epsilon_k G_k)$ with $\|G_k\| \leq 1$, Proposition 5 gives a first-order bound that does not depend on the direction,

$$|\Delta E_k| \leq 2\sigma_H \epsilon_k + O(\epsilon_k^2). \quad (20)$$

The allocator uses $g_k = \max(|\partial E/\partial \theta_k|, 2\sigma_H)$ in (18). Three consequences follow. On an eigenstate the bound is zero and the quadratic model of Proposition 4 remains. On a state with residual ansatz error the first-order term dominates the on-axis curvature for every ϵ_k in the working range, so the tolerable synthesis error scales as δ_{syn} and not as $\sqrt{\delta_{\text{syn}}}$. And because the bound does not distinguish rotations, a per-rotation allocation of precision needs either an eigenstate, or the actual error generator of the synthesis algorithm, or mixing-based synthesis [20, 21], which turns the coherent error into a stochastic Pauli-type error of probability of order ϵ_k^2 on S_k that the sensitivity tensor of Section 4 already describes. The cap $\epsilon_k \leq 10^{-2}$ is kept, and the second-order cross terms between rotations are omitted.

7.3 Allocation and closed forms

The synthesis allocation problem is

$$\min_{\epsilon} \sum_k a \log_2(1/\epsilon_k) \quad \text{subject to} \quad \sum_k \Delta E_k(\epsilon_k) \leq \delta_{\text{syn}}. \quad (21)$$

Its stationarity condition is $a/(\epsilon_k \ln 2) = \lambda(g_k + h_k \epsilon_k)$, a quadratic in ϵ_k with one multiplier λ found by bisection. With $h = 0$ the solution is $\epsilon_k = \delta_{\text{syn}}/(K g_k)$ and the saving relative to a uniform precision is $a K \log_2[\text{AM}(g)/\text{GM}(g)]$, with AM and GM the arithmetic and geometric means over rotations. With $g = 0$ the solution is $\epsilon_k \propto h_k^{-1/2}$ and the saving is $\frac{a}{2} K \log_2[\text{AM}(h)/\text{GM}(h)]$. In both cases the saving is set by how unevenly the sensitivities are spread. The continuous solution is rounded to integers by (17), and when rounding makes it worse than the uniform precision the uniform precision is kept.

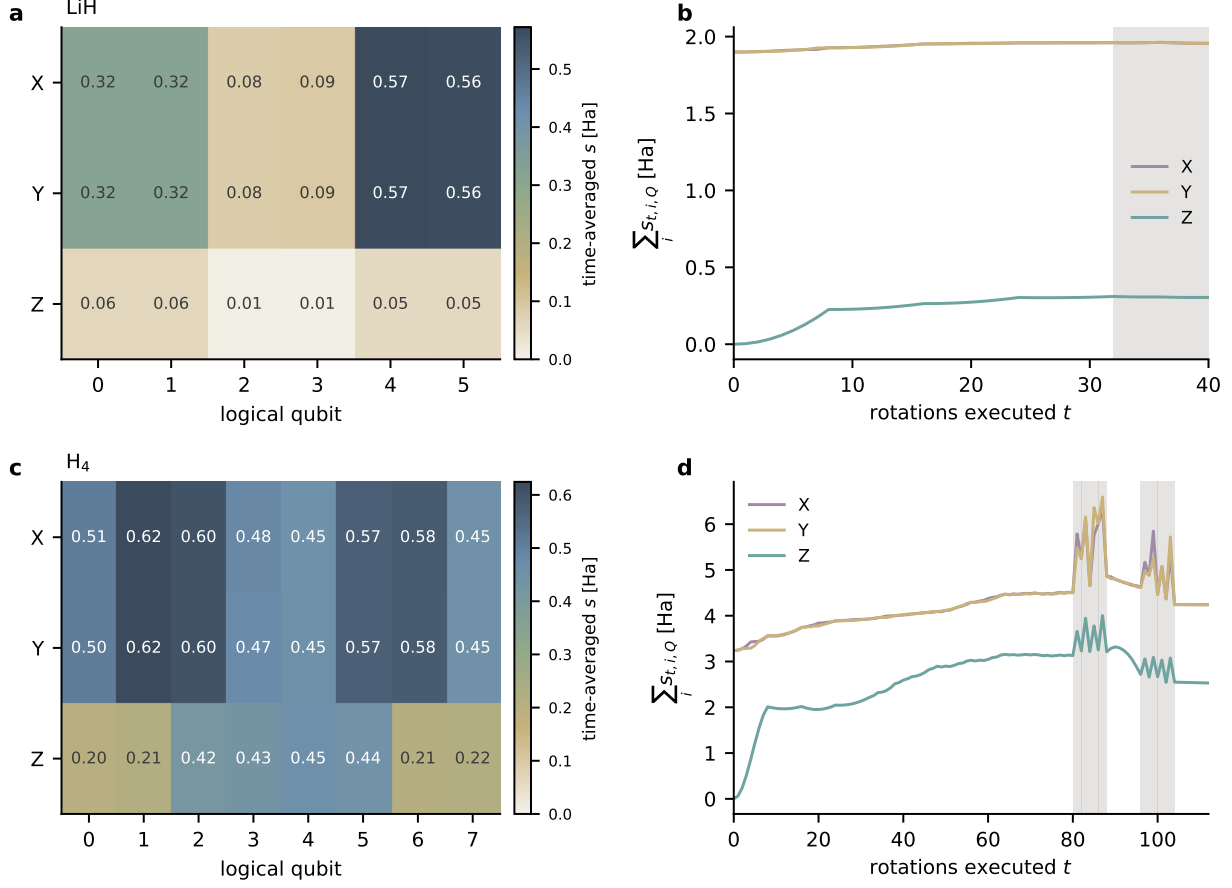


Figure 2: Sensitivity of the energy to Pauli errors. (a, c) Time-averaged sensitivity by logical qubit and Pauli type for LiH and H₄. (b, d) The sum over qubits as a function of the number of rotations executed before the error, by Pauli type. Grey bands mark the rotations that belong to single-excitation generators; the other rotations belong to double excitations.

7.4 Budget split

The fraction $f = \delta_{\text{mem}} / (\delta_{\text{mem}} + \delta_{\text{syn}})$ is scanned over a grid. For each value δ_T is computed, the synthesis problem is solved, its n_T fixes the durations (11), the distance problem is solved, and the split with the smallest V_{total} is kept.

8 Findings on two systems

Both inputs are generated with PySCF [22] in the STO-3G basis. The first is LiH at $R = 1.6 \text{ \AA}$ with the Li 1s core frozen and the three valence orbitals of A₁ symmetry active, $n = 6$ and two electrons. ADAPT-VQE converges in 8 generators, 40 rotations, to the exact active-space energy within 10^{-14} Ha. The second is an H₄ chain with $1.5 \text{ \AA}$ spacing and all four orbitals active, $n = 8$ and four electrons, strongly correlated with 0.167 Ha of correlation energy. ADAPT-VQE converges in 20 generators, 112 rotations, within 10^{-14} Ha. In both cases the active-space Hamiltonian reproduces the complete active space configuration interaction (CASCI) energy of PySCF to 10^{-8} Ha in the sector of N electrons and $S_z = 0$.

8.1 LiH

Table 1 lists the X and Z sensitivities of an error at the end of the circuit next to the orbital energies, and Figure 2a shows the time average. The X sensitivities exceed $|\varepsilon_i|$ by 0.008 to 0.034 Ha, the size of the correlation energy, 0.019 Ha, as Proposition 3 predicts for a weakly correlated system. The Z sensitivities are smaller by a factor of 5 to 17, as Proposition 2 predicts, and they are not uniformly small. They are largest on the 2σ and 4σ orbitals, between which the dominant double excitation acts, and smallest on the 3σ orbital. Summed over the tensor the ratio of X to Z sensitivity is 8.2.

qubits	orbital	n_i	ε_i [Ha]	$s_{X,i}$ [Ha]	$s_{Z,i}$ [Ha]
0, 1	2σ (HOMO)	0.977	-0.285	0.319, 0.315	0.076
2, 3	3σ (LUMO)	0.006	+0.078	0.086, 0.098	0.011
4, 5	4σ	0.017	+0.548	0.573, 0.565	0.066

Table 1: LiH. Occupation n_i of the spin-orbital in the exact ground state, orbital energy, and the sensitivity of the energy to a single X or Z error at the end of the circuit. HOMO and LUMO are the highest occupied and lowest unoccupied molecular orbitals. The two X values of a pair are the two spin-orbitals; the Z values of a pair are equal.

Figure 2b shows the time profile. The X sum is constant to 3% across the circuit. The Z sum is exactly zero at $t = 0$, where the state is the determinant, and rises to its final value within the first 25 rotations as the circuit builds the correlation. The concentration (10) is 0.001 along the time axis, 0.26 along the qubit axis, 0.26 along the type axis and 0.46 over the joint (qubit, type) axis. Collapsing the type index by a maximum over Q reduces the joint concentration to 0.27. Figure 3a places the X sensitivities against the orbital energies, and Figure 3b places the Z sensitivities against the bound (9). The bound holds at every qubit, with a slack of 9 to 23 on LiH. A Cauchy–Schwarz bound summed over many terms is loose by factors of this size.

Proposition 5 was checked numerically on this system (Figure 4). The directions G are the eight excitation generators of the ADAPT pool, normalised to $\|G\| = 1$, each inserted at the end of the circuit and at two random positions. On the converged circuit σ_H is zero to machine precision, every first-order coefficient is below 2×10^{-8} Ha, and the shift scales as $\varepsilon^{2.00}$ for all 24 insertions. On the ansatz truncated to three generators, residual error 9.6×10^{-4} Ha, one has $\sigma_H = 2.07 \times 10^{-2}$ Ha, the bound $2\sigma_H = 4.13 \times 10^{-2}$ Ha, and the largest first-order coefficient 3.99×10^{-2} Ha. On the ansatz truncated to one generator, residual error 4.6×10^{-3} Ha, the bound is 0.108 Ha and the largest coefficient 0.074 Ha. Directions that are already in the ansatz keep the quadratic scaling, as Proposition 4 requires. A residual error of 10^{-3} Ha costs about $0.04 \epsilon_k$ per rotation at first order, which exceeds the curvature term $h_k \epsilon_k^2 / 2$ for every ϵ_k below 5×10^{-2} . For a synthesis budget of 10^{-3} Ha over 40 rotations this means $\epsilon_k \approx 10^{-2}$ and about 20 magic states per rotation on the converged circuit, against $\epsilon_k \approx 6 \times 10^{-4}$ and about 32 on the truncated one.

8.2 H₄ chain

The ratio of X to Z sensitivity drops to 1.65 (Figure 2c). The X sensitivities lie between 0.31 and 0.72 Ha at $t = K$ and do not follow the orbital energies (Figure 3a); the orbital with the smallest $|\varepsilon_i|$, 0.133 Ha, has occupation 0.18 in the ground state and the Koopmans relation, a property of the determinant, no longer applies. The Z sensitivities take two values, 0.26 to 0.27 Ha on the outer orbitals and 0.36 to 0.37 Ha on the two frontier orbitals, and the ratio $s_{Z,i} / \sqrt{n_i(1 - n_i)}$ is 0.94 to 0.97 on all four orbitals. The bound (9) holds with a slack of 14 to 15. The concentration is

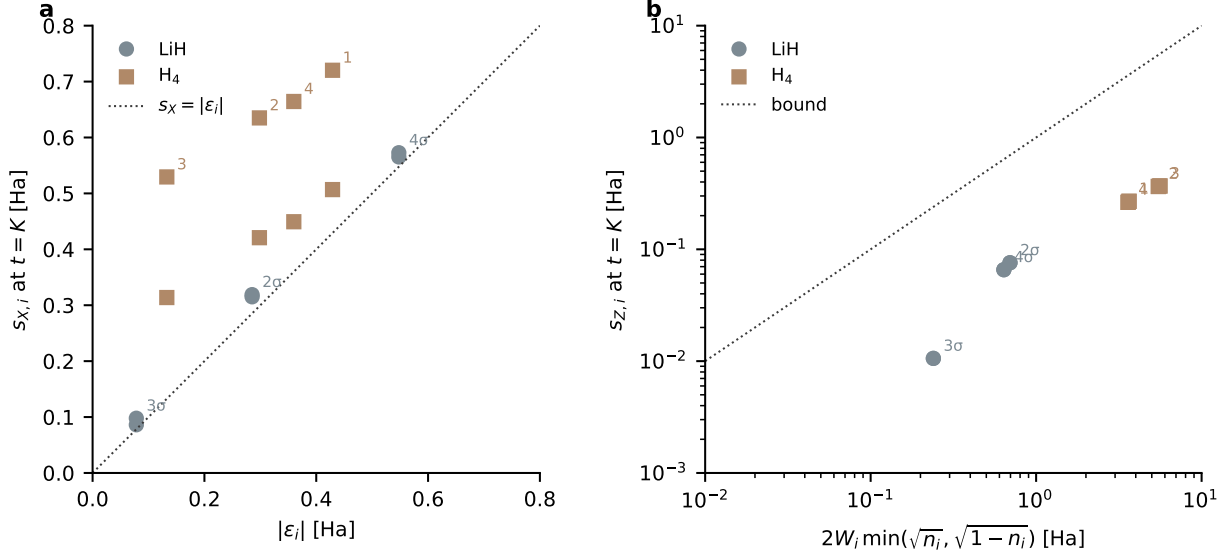


Figure 3: Structure of the tensor at $t = K$. (a) X sensitivity of each spin-orbital against the magnitude of its orbital energy; the dotted line is the mean-field relation of Proposition 3. (b) Z sensitivity of each spin-orbital against the bound (9); the dotted line is equality. Labels give the orbital.

0.013 along the qubit axis and 0.073 over the joint axis. A phase error costs almost as much as an occupation error, and per-qubit allocation has nothing to work with.

The propositions predict this. The asymmetry that sensitivity-based allocation exploits is the mean-field part of the energy. It is largest for weakly correlated systems and it shrinks with correlation strength. The concentration η shows it before any allocation is attempted. Whether the structure returns in the large active spaces that fault-tolerant chemistry targets, where most orbitals are weakly correlated and a few carry the strong correlation, is the first thing a larger-scale sensitivity method should check.

Two features of Figure 2d are not predicted by Propositions 2 and 3 and are explained by the encoding. First, the time profile has transients of up to 3 Ha in the sum over qubits, and they coincide with the single-excitation generators. A fermionic generator maps to a set of commuting Pauli strings that conserve the particle number only as a group. After part of the group has been applied the state has a component outside the N -electron sector, and an error at that moment prevents the remaining strings from removing it, so the corrupted output carries an $N \pm 2$ component whose energy is far from the ground state. The effect grows with the angle of the string just executed. Over the 92 interior slices of the H_4 circuit the excess of the sum over the two enclosing generator boundaries has correlation 0.91 with $\sin^2$ of that angle, with slope 6.2 Ha per unit of $\sin^2$; the mean excess is 2.2 Ha inside single excitations, whose two strings carry angle $\theta/2$, and -0.04 Ha inside double excitations, whose eight strings carry $\theta/8$. The largest string angle is 0.75 rad. On LiH every angle is below 0.02 rad and the profile is flat. The order in which the strings of a generator are executed is a compilation choice, and the transient is a cost of that choice that the tensor makes visible.

Second, the two spin-orbitals of a spatial orbital have different X sensitivities, 0.51 against 0.72 Ha on the first orbital, while their Z sensitivities are equal to 10^{-8} Ha. Rebuilding the Hamiltonian and the circuit with the β spin-orbital of each pair on the even qubit reproduces the energy to

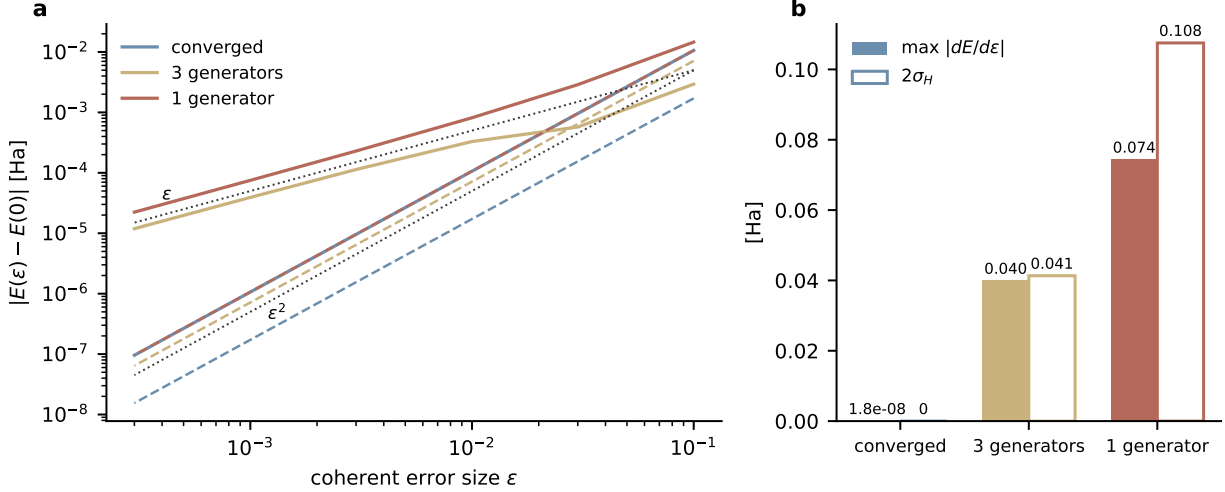


Figure 4: Coherent errors on LiH. (a) Energy shift against the size of a coherent error $\exp(-i\epsilon G)$ for the converged circuit and for the ansatz truncated to three and to one generator. For each circuit the solid line is the insertion with the largest first-order coefficient and the dashed line the one with the smallest. Dotted references scale as ϵ and ϵ^2 . (b) Largest first-order coefficient over the 24 insertions against the bound $2\sigma_H$ of Proposition 5, with values above the bars. On the converged circuit the coefficient is 1.8×10^{-8} Ha and the bound is zero.

10^{-15} Ha and leaves every sensitivity unchanged qubit by qubit. The asymmetry therefore belongs to the position of the qubit in the ordering, not to the spin, which is Remark 1. On LiH the partner occupations are within 0.03 of 0 or 1 and the two values of a pair agree to 1%.

8.3 Allocation

Table 2 collects the allocation at chemical accuracy with $p = 10^{-3}$, $D = \{3, 5, \dots, 33\}$ and the cost parameters of Section 6. The magic-state term δ_T is 2.0×10^{-5} Ha for LiH and 8.7×10^{-5} Ha for H_4 , 1 to 5% of the budget. On both circuits σ_H is zero to machine precision, so the bound (20) does not act. On both molecules and both observables the sensitivity-aware allocation returns the uniform distances. The gap rule (16) explains this. The largest sensitivity ratio in LiH, between the 4σ and 3σ orbitals, is 6.5, below $\Lambda = 8.7$, and gives a gap of 1.7 where a step is 2. In H_4 the ratio is 1.4 and the gap 0.3. The mean-field control, which sets the Z sensitivities to zero, proposes a smaller distance on one LiH qubit and on every H_4 qubit and is infeasible on the true tensor in both cases. The Z sensitivities, one tenth of the total on LiH, are needed. The shuffled control is uninformative while the allocation is uniform.

system	observable	d uniform	d sensitivity-aware	d type-agnostic	p_L required
LiH	energy	13	all 13	all 13	2.5×10^{-7}
LiH	overlap	13	all 13	all 15	
H_4	energy	15	all 15	all 15	2.3×10^{-8}
H_4	overlap	15	all 15	all 15	

Table 2: Allocation at chemical accuracy and $p = 10^{-3}$. The last column is the per-patch, per-round logical error rate that uniform protection needs for the energy at chemical accuracy.

Type resolution changes one entry of the table. For the overlap observable on LiH the type-agnostic allocation, which assigns the large occupation sensitivity to the phase channel as well, needs $d = 15$ on every patch, 1350 physical data qubits against 1014 at $d = 13$. Resolving the type avoids a third more qubits without changing the accuracy.

The last column of the table is the logical error rate per patch and per round that uniform protection needs, 2.5×10^{-7} for LiH and 2.3×10^{-8} for H_4 . A $\pi/8$ rotation occupies 13 rounds at $d = 13$, so the LiH figure is about 3×10^{-6} per patch and per logical operation over about 900 logical operations. This is the number to compare with the logical error rate of a given machine.

Savings of 20 to 40% were anticipated at the outset. They do not appear on these two molecules, for three reasons. The sensitivity spread is below Λ . The total volume is dominated by the magic-state factories, 91% of V_{total} at the LiH uniform solution, so a change of distance on a few patches moves the total by about one percent even where it is allowed. And the merge rule (12) is conservative, since a small patch drags every measurement it participates in down to its own distance, and LiH rotations involve three to six of the six qubits.

9 Assumptions and limitations

The error functional (13) is first order in the error probabilities, and correlated failures and coherent accumulation are not modelled. Errors are stochastic Paulis; leakage, crosstalk and loss are absent from the noise model, only the uniform noise structure is reported, and logical Y errors are dropped. During a multi-patch measurement the protection of the participants is taken as the smallest participating distance and the duration as the largest, which are first-order approximations to the lattice-surgery cost, the first of them conservative. Errors in the routing region and in the factories enter only through δ_T and c_T . The time model assumes lattice surgery on a planar surface-code layout.

The logical error model (4) is fitted at $d \leq 7$ and used at $d \leq 33$; the hold-out test bounds the risk of the extrapolation without removing it. Sensitivities are computed by state-vector simulation, exact for $n \leq 12$. The energy is treated as an expectation value, and for phase estimation the overlap tensor is the relevant one. Synthesis error is modelled by the direction-independent first-order bound (20) and the on-axis curvature, with $\epsilon_k \leq 10^{-2}$. Two molecules of six and eight qubits demonstrate the structure of the tensor and are not a benchmark.

10 Next steps

Derive the general expression of $s_{X,i}$ in terms of the one- and two-body density matrices, with Proposition 3 as its mean-field limit. Follow LiH and H_4 along a bond-length scan and test the $\sqrt{n_i(1-n_i)}$ scaling of $s_{Z,i}$ orbital by orbital. Treat LiH in its full STO-3G space, twelve qubits, where the core and the π orbitals give sensitivity ratios well above Λ . Validate the first-order functional (13) against direct sampling at elevated error rates. Order the strings of a generator, or execute a generator as one block, to remove the transients of Section 8.2. Replace the merge rule (12) by a model in which only the measurement outcome is exposed at the smaller distance. Solve the synthesis problem over integer T counts and implement the mixing variant. Add a neutral-atom architecture module with transversal gates and atom transport. Replace state-vector sensitivities by Heisenberg-picture Pauli propagation with truncation for circuits of 30 to 60 logical qubits.

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